



BAULKHAM HILLS HIGH SCHOOL

2016

**YEAR 12 HSC ASSESSMENT TASK 2
HALF YEARLY**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 11-14
- Marks may be deducted for careless or badly arranged work

Total marks – 70

Exam consists of 9 pages.

This paper consists of TWO sections.

Section 1 – Pages 2-4 (10 marks)

Questions 1-10

- Attempt Questions 1-10
Allow about 15 minutes for this section.

Section II – Pages 5-9 (60 marks)

- Attempt questions 11-14
Allow about 1 hour and 45 minutes for this section.

Reference Sheet is provided.

Section I - Multiple Choice (10 marks)

Allow about 15 minutes for this section.

Use the multiple choice page for Question 1-10

1 The remainder when $P(x) = 2x^3 - 5x - 3$ is divided by $(2x + 1)$?

(A) -6

(B) $-\frac{3}{4}$

(C) 0

(D) $\frac{3}{4}$

2 What are the solutions to the equation

$$e^{6x} - 7e^{3x} + 6 = 0$$

(A) $x = 1$ and $x = 6$

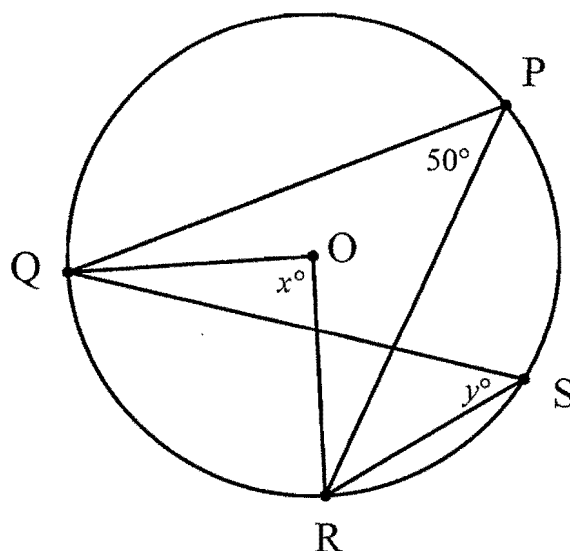
(B) $x = 0$ and $x = \frac{\log_e 6}{2}$

(C) $x = 0$ and $x = \frac{\log_e 6}{3}$

(D) $x = 1$ and $x = \frac{\log_e 6}{2}$

3 P, Q, R and S are points on a circle with centre O .

$\angle QPR = 50^\circ$. Find the values of $x = \angle QOR$ and $y = \angle QSR$



(A) $x = 50^\circ, y = 50^\circ$

(B) $x = 100^\circ, y = 25^\circ$

(C) $x = 100^\circ, y = 50^\circ$

(D) $x = 50^\circ, y = 25^\circ$

- 4 $\int 6 \cos^2 3x \, dx =$
- (A) $2 \sin^3 3x + c$
- (B) $2 \cos^3 6x + c$
- (C) $3x - \frac{1}{2} \sin 6x + c$
- (D) $3x + \frac{1}{2} \sin 6x + c$
- 5 What is the solution to
- $$\frac{2t+1}{t-2} > 1$$
- (A) $t > -3$
- (B) $t < -3$ or $t > 2$
- (C) $-3 < t < 2$
- (D) $t < -2$ or $t > 3$
- 6 The number of ways can a committee of 12 be chosen from 15 men and 6 women so that there are at least 11 men is
- (A) ${}^{21}C_{12}$
- (B) ${}^{15}C_{11} \times {}^{12}C_1$
- (C) ${}^{15}C_{12} + {}^{15}C_{11} \times 6$
- (D) ${}^{15}C_{12} + {}^{15}C_{11}$
- 7 Find the inverse function, $f^{-1}(x)$, in terms of x , when the function $f(x)$ is given by
- $$y = \frac{2}{x+1}$$
- (A) $y = \frac{2-x}{x}$
- (B) $y = \frac{x+1}{2}$
- (C) $y = \frac{2-x}{2}$
- (D) $y = \frac{2+x}{x}$

- 8 If x is an acute angle, which of the following is equivalent to the expression

$$\sqrt{\frac{4 + 4 \cos 2x}{1 - \cos 2x}}$$

- (A) $2 \cot^2 x$
(B) $2 \cot x$
(C) $2 \tan x$
(D) $2 \tan^2 x$
- 9 A point $P(x, y)$ moves such that $x = \sin \theta$, $y = \operatorname{cosec} \theta$.
Which of the following best describes the locus of P ?
- (A) Parabola
(B) Circle
(C) Hyperbola
(D) Straight Line
- 10 Find $f'(x)$ if $f(x) = \tan^{-1}(x) + x \tan^{-1}(x)$
- (A) $\frac{1}{1+x^2}$
(B) $\frac{x+1}{1+x^2}$
(C) $\tan^{-1} x + \frac{1}{1+x^2}$
(D) $\tan^{-1} x + \frac{x+1}{1+x^2}$

End of Section 1

Section II (90 marks)**Allow about 2 hours and 45 minutes for this section.****Answer each question on the appropriate page in the writing booklet.****Question 11 (15 marks)****Marks**

- a) Show that

$$\lim_{x \rightarrow 0} \frac{\sin 5x}{7x} = \frac{5}{7}$$

1

- b) Given
- $y = 3 \cos^{-1} 2x$

- (i) State the domain and range

2

- (ii) Hence sketch
- $y = 3 \cos^{-1} 2x$

2

- c) An object, removed from a freezer at
- -5°C
- , is placed in a room where the temperature is kept constant at
- 15°C
- . Thereafter, the object's temperature,
- $T^{\circ}\text{C}$
- , is changing so that after
- t
- minutes,

$$\frac{dT}{dt} = k(15 - T) \quad \text{where } k \text{ is constant}$$

- (i) Verify that the function
- $T = 15 - Ae^{-kt}$
- satisfies this condition

1

- (ii) Find the value of
- A

1

- (iii) If initially, the temperature was increasing at
- 5°C
- per minute, find the value of
- k

1

- (iv) Find to the nearest second, the time taken for the temperature to rise to
- 0°C

1

- d) Find the acute angle (correct to the nearest minute) between the lines

2

$$2x - 3y - 1 = 0 \text{ and } y = \frac{x}{5} - 7.$$

- e) A curve is defined by the parametric equations
- $x = t - 3$
- and
- $y = t^2 - 9$
- .

- (i) Show that
- $\frac{dy}{dx} = 2x + 6$
- .

2

- (ii) Find the equation of the normal to the curve at
- $t = -3$
- .

2**End of Question 11**

Question 12 (15 marks)

a) Given $y = 1 + \sin 4t + \sqrt{3} \cos 4t$:

(i) Express $\sin 4t + \sqrt{3} \cos 4t$ in the form $R \sin(4t + \alpha)$. 2

(ii) Hence or otherwise, solve $1 + \sin 4t + \sqrt{3} \cos 4t = 0$ for the domain $0 \leq t \leq \frac{\pi}{2}$. 2

b) A polynomial $P(x)$ is monic and of degree 5.

It has a single zero at $x = -2$ and a double zero at $x = 1$. 3

The other two zeroes are equal in magnitude and opposite in sign.

Given $P(0) = -18$, write $P(x)$ in fully factorised form.

c) Evaluate

$$\int_1^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}}$$

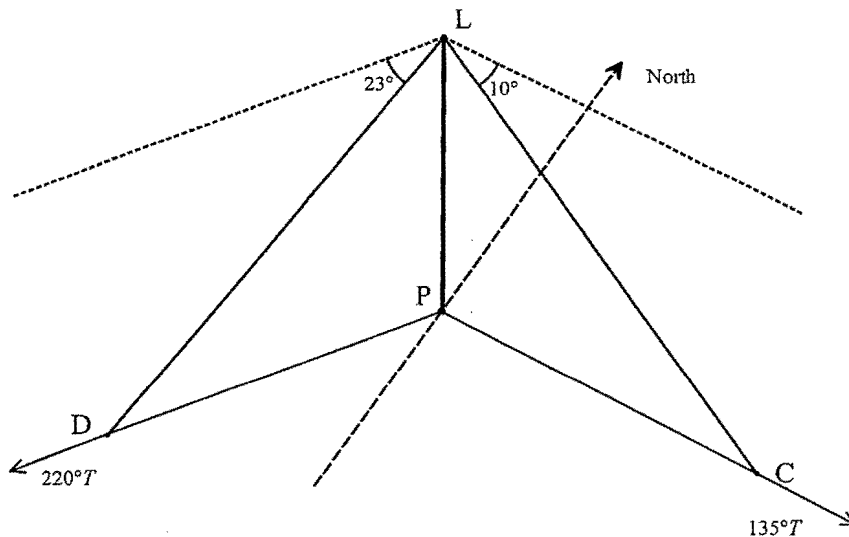
2

d) Solve for x

$$x + 4 \geq \frac{2}{x + 3}$$

3

e) From the base of a lighthouse (P), a yacht (C) is on a bearing of $135^\circ T$ and another vessel (D) is on a bearing of $220^\circ T$. From the top of the lighthouse (L), 115 metres above sea-level, Yacht C is at an angle of depression of 10° while vessel D is at an angle of depression of 23° . 3



Calculate the distance between the two vessels at that moment, correct to the nearest metre.

End of Question 12

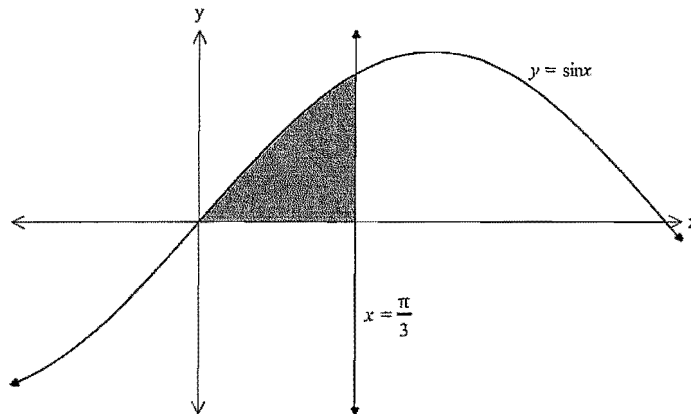
Question 13 (15 marks)**Marks**

- a) (i) Show that $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$ **1**
- (ii) Hence or otherwise, find the exact value of **2**
- $$\int_0^{\frac{\pi}{6}} \sin 4x \cos 2x \, dx$$
- b) Use the method of mathematical induction to prove that $4^n + 14$ is a multiple of 6 for all integers $n \geq 1$. **3**
- c) The equation $x^3 + 2x^2 + 3x + 6 = 0$ has roots α, β and γ . **2**
- Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$
- d) Consider the word **M A T H E M A T I C S**
- (i) Find how many distinct ways the letters can be arranged **1**
- (ii) How many arrangements are the 4 vowels together? **2**
- (iii) How many arrangements begin and end with the letter **M**? **2**
- e) Show that $\sin\left(2 \cos^{-1}\left(\frac{p-q}{p+q}\right)\right) = \frac{4\sqrt{pq}(p-q)}{(p+q)^2}$ **2**

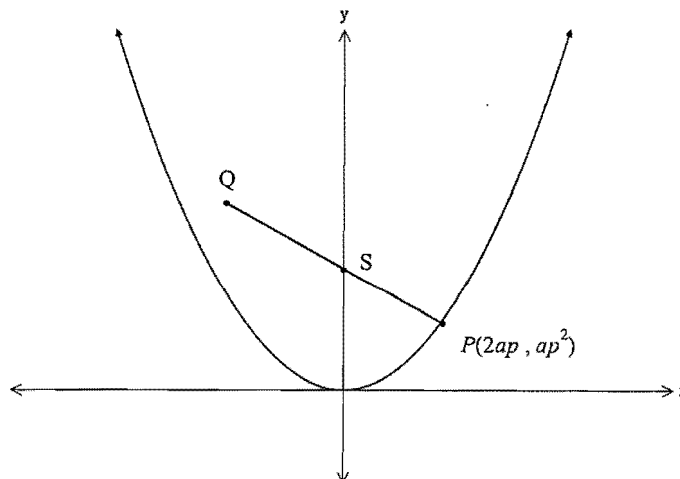
End of Question 13

Question 14 (15 marks)

- a) Find the volume formed when the area bounded by the curve $y = \sin x$, the line $x = \frac{\pi}{3}$ and the x -axis is rotated about the x -axis. 3



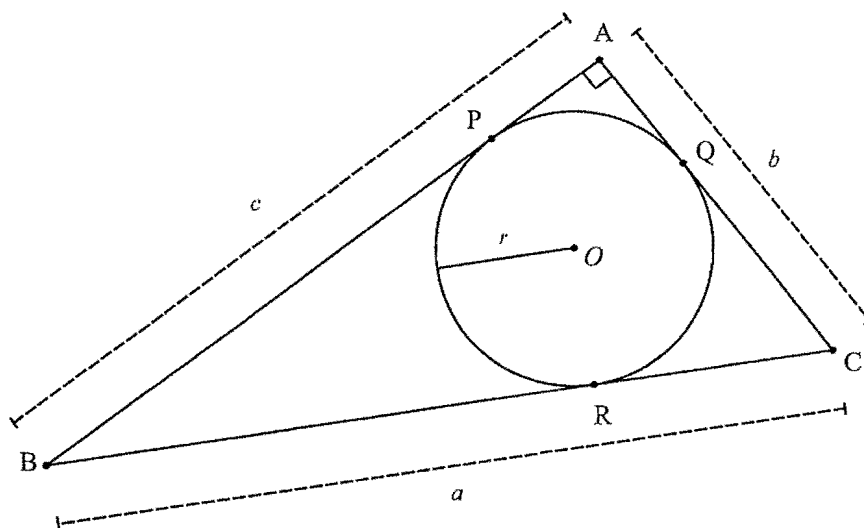
- b) After t months, the number of goats on an island is given by $N = 500 - 400e^{-0.1t}$.
- Sketch the graph of N as a function of t , showing clearly the initial population size. 2
 - Show that the rate of growth $\frac{dN}{dt}$ is given by $\frac{dN}{dt} = 0.1(500 - N)$. 1
 - Find the population size for which the rate of growth is half the initial rate of growth. 1
- c) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$ with focus, $S(0, a)$. The point Q lies on PS produced and Q divides PS externally in the ratio 4:3.



- Show that Q has co-ordinates $(-6ap, a(4 - 3p^2))$ 2
- Show that the locus of Q , as P varies, is a new parabola and find its focal point. 2

Question 14 continued on the next page

- d) A triangle ABC , is right angled at A and has sides of length a, b and c . A circle of radius r , centre at O , is drawn so that the sides of the triangle are tangents to the circle. P, Q and R are the points where the tangents AB, AC and BC meet the circles respectively.



Copy the diagram into your answer booklet

- i) Prove that $AQOP$ is a square

2

- ii) Prove that PA is $\frac{c+b-a}{2}$

2

End of Exam

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \tan\phi}$$

t formulae

If $t = \tan \frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

tangent: $y = tx - at^2$

normal: $x + ty = at^3 + 2at$

At (x_1, y_1) ,

tangent: $xx_1 = 2a(y + y_1)$

normal: $y - y_1 = -\frac{2a}{x_1}(x - x_1)$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

REFERENCE SHEET

— Mathematics —

— Mathematics Extension 1 —

— Mathematics Extension 2 —

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

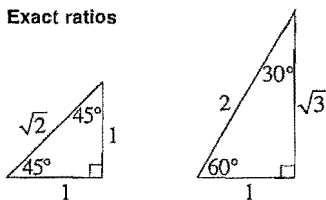
$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

$$\text{If } y = x^n, \text{ then } \frac{dy}{dx} = nx^{n-1}$$

$$\text{If } y = uv, \text{ then } \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\text{If } y = \frac{u}{v}, \text{ then } \frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$\text{If } y = F(u), \text{ then } \frac{dy}{dx} = F'(u) \frac{du}{dx}$$

$$\text{If } y = e^{f(x)}, \text{ then } \frac{dy}{dx} = f'(x) e^{f(x)}$$

$$\text{If } y = \log_e f(x) = \ln f(x), \text{ then } \frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$\text{If } y = \sin f(x), \text{ then } \frac{dy}{dx} = f'(x) \cos f(x)$$

$$\text{If } y = \cos f(x), \text{ then } \frac{dy}{dx} = -f'(x) \sin f(x)$$

$$\text{If } y = \tan f(x), \text{ then } \frac{dy}{dx} = f'(x) \sec^2 f(x)$$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x - h)^2 = \pm 4a(y - k)$$

Integrals

$$\int (ax + b)^n dx = \frac{(ax + b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + C$$

$$\int \cos(ax + b) dx = \frac{1}{a} \sin(ax + b) + C$$

$$\int \sec^2(ax + b) dx = \frac{1}{a} \tan(ax + b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Multiple Choice

Q1 $P(-\frac{1}{2}) = 2(-\frac{1}{2})^3 - 5(-\frac{1}{2}) - 3$
 $= -0.75$ $\therefore$ (B)

$$= 2x \sqrt{\frac{1+t^2+1-t^2}{1+t^2-1-t^2}}$$

$$= 2x \sqrt{\frac{2}{-2t^2}} = 2\sqrt{\frac{1}{t^2}} = 2\frac{1}{t}$$

$$= 2 \cot x$$
 (B)

Q2. $(e^{3x} - 6)(e^{3x} - 1) = 0$
 $e^{3x} = 6$ or $e^{3x} = 1$

$x = \frac{\ln 6}{3}$ or $x = 0$ $\therefore$ (C) Q9. $x = \sin \theta$, $y = \frac{1}{\sin \theta}$

$\therefore y = \frac{1}{x}$ (C)

Q3. $x = 100$ (Lat centre = $2x$ Lat circumference)

$y = 50$ ($\angle$ standing on same arc) $\therefore$ (C)

Q10 $f(x) = \tan^{-1} x + x + \tan^{-1} x$

$f'(x) = \frac{1}{x^2+1} + \tan^{-1} x + \frac{1}{x^2+1} \cdot x$
 $= \tan^{-1} x + \frac{x+1}{x^2+1}$ (D)

Q4. $\int 6 \times \frac{1}{2} (\cos 6x + 1) dx$
 $= \int 3 \cos 6x + 3 dx$
 $= \frac{1}{2} \sin 6x + 3x + C$ (D)

Q5. $t \neq 2$

boundary: $2t+1 = t-2$

$t = -3$ (B)

test $t=0$ $\frac{1}{-2} \neq 1$

Q6. ${}^{15}C_{12}$ if all men.

${}^{15}C_{11} \times 6$ for 11 men and 1 woman. (C)

Q7. inverse: $x = \frac{2}{y+1}$

$y+1 = \frac{2}{x}$

$y = \frac{2}{x} - 1$ (A)

Q8 $\int \frac{4(1 + \cos 2x)}{1 - \cos 2x}$

$= 2x \sqrt{\frac{1 + \frac{1-t^2}{1+t^2}}{1 - \frac{1-t^2}{1+t^2}}}$ where $t = \tan x$

Question 11.

marks.

a) $\lim_{x \rightarrow 0} \frac{5x \sin 5x}{7x} = \lim_{x \rightarrow 0} \frac{5}{7} \frac{\sin 5x}{5x}$
 $= \frac{5}{7}$

1mk - correct soln.

b) $y = 3 \cos^{-1} 2x$

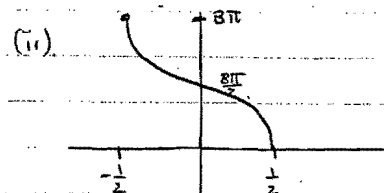
(i) Domain: $-\frac{1}{2} \leq x \leq \frac{1}{2}$

Range: $0 \leq y \leq 3\pi$

2mk - correct answer

either correct domain

OR range.



2mk - correct graph

and relevant values

1mk either correct

graph or correct values

c) (i) $T = 15 - Ae^{-kt} \rightarrow -Ae^{-kt} = T - 15$

1mk - correct soln

$\frac{dT}{dt} = -kx - Ae^{-kt}$
 $= -k(T - 15)$
 $= k(15 - T)$

(ii) when $t=0$, $T = -5^\circ C$

$-5 = 15 - Ae^0$

$A = 20$

1mk - correct answer

(iii) when $t=0$ $\frac{dT}{dt} = 5$

1mk - correct

answer.

$5 = k(15 - (-5))$

$k = 0.25$

(iv) $0 = 15 - 20e^{-0.25t}$

1mk correct

answer

$\frac{15}{20} = e^{-0.25t}$
 $-0.25t = \ln\left(\frac{15}{20}\right)$

$t = 1.15$

ie 1min and 9 sec.

11. d) $3y = 2x - 1$ $y = \frac{x}{3} - \frac{1}{3}$
 $y = \frac{2}{3}x - \frac{1}{3}$ $m_2 = \frac{1}{3}$
 $m_1 = \frac{2}{3}$

$$\tan \theta = \left| \frac{\frac{2}{3} - \frac{1}{3}}{1 + \frac{2}{3} \times \frac{1}{3}} \right|$$

$$= \left| \frac{1}{7} \right|$$

$$\theta = 22^\circ 23' \text{ (nearest min)}$$

$$\text{OR } \theta = 22.38^\circ$$

marks.

2mk - correct soln

1mk - correct use

of formula.

• finds both gradients.

e) (i) $x = t - 3$ $y = t^2 - 9$ OR (method 2)
 $t = x + 3$
 $y = (x + 3)^2 - 9$
 $y = x^2 + 6x$
 $\frac{dy}{dx} = 2x + 6$
 $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$
 $\frac{dx}{dt} = 1$ $\frac{dy}{dt} = 2t$
 $\therefore \frac{dy}{dx} = 2t$
 $\frac{dy}{dx} = 2(x + 3)$

2mk - correct soln.

1mk - eliminating t

• correct

derivative

(ii) at $t = -3$
 $x = -6$, $y = 0$.

$$m_T = -12 + 6$$

$$= -6$$

$$m_N = \frac{1}{6}$$

eqn of normal:

$$y - 0 = \frac{1}{6}(x + 6)$$

$$x - 6y + 6 = 0.$$

2mk - correct soln.

1mk - finding gradient of normal.

• finding x & y values and

substituting into eqn of the line.

Question 12

a) $y = 1 + \sin 4t + \sqrt{3} \cos 4t$

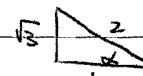
(i) $\sin 4t + \sqrt{3} \cos 4t = R \sin(4t + \alpha)$
 $= R \sin 4t \cos \alpha + R \cos 4t \sin \alpha$

$$R \cos \alpha = 1$$

$$R \sin \alpha = \sqrt{3}$$

$$\tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$



$$\therefore R = 2$$

$$\therefore \sin 4t + \sqrt{3} \cos 4t = 2 \sin(4t + \frac{\pi}{3})$$

(ii) $1 + \sin 4t + \sqrt{3} \cos 4t = 0$ $0 \leq t \leq \frac{\pi}{2}$
 $2 \sin(4t + \frac{\pi}{3}) = -1$ $\frac{\pi}{3} \leq 4t + \frac{\pi}{3} \leq 2\pi + \frac{\pi}{3}$

$$\therefore 4t + \frac{\pi}{3} = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$4t = \frac{5\pi}{6}, \frac{9\pi}{2}$$

$$t = \frac{5\pi}{24}, \frac{9\pi}{8}$$

2mk - correct

R and α

1mk - finds R

• finds α

2mk - correct soln

1mk

• finds correct

angle for

$$(4t + \frac{\pi}{3})$$

b) (a) $P(x)$ monic, deg of 5 $\rightarrow x^5$

(b) zero at $x = -2 \rightarrow (x + 2)$

(c) double zero at $x = 1 \rightarrow (x - 1)^2$

(d) equal in mag and opp in sign $\rightarrow (x - \alpha)(x + \alpha)$

$$P(x) = (x + 2)(x - 1)^2(x - \alpha)(x + \alpha)$$

$$P(0) = -18$$

$$-18 = (2)(-1)^2(0 - \alpha^2)$$

$$+9 = +\alpha^2$$

$$\alpha = \pm 3$$

$$\therefore P(x) = (x + 2)(x - 1)^2(x - 3)(x + 3)$$

3mk - correct soln

2mk satisfies

any 3 of ABCD

1mk satisfies

any 2 of ABCD

c) $\int_1^{\sqrt{3}} \frac{dx}{1 + x^2} = \left[\sin^{-1} \frac{x}{2} \right]_1^{\sqrt{3}}$

$$= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{1}{2}$$

$$= \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$$

2mk correct soln

1mk

• correct integration

• correct evaluation

Q12d) $x+4 \geq \frac{2}{x+3}$

multiply both sides by $(x+3)^2$

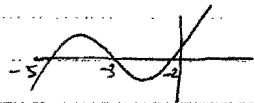
$$(x+4)(x+3)^2 \geq 2(x+3)$$

$$(x+4)(x+3)^2 - 2(x+3) \geq 0$$

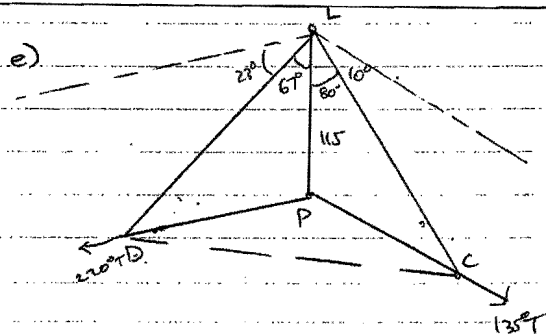
$$(x+3)[(x+4)(x+3) - 2] \geq 0$$

$$(x+3)(x^2+7x+10) \geq 0$$

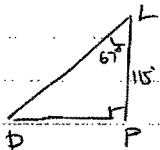
$$(x+3)(x+2)(x+5) \geq 0$$



$$-5 \leq x < -3 \text{ or } x \geq -2$$



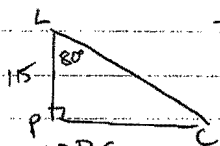
In $\triangle LPD$



$$\tan 67^\circ = \frac{DP}{115}$$

$$DP = 115 \tan 67^\circ$$

In $\triangle LPC$



$$\tan 80^\circ = \frac{PC}{115}$$

$$\therefore PC = 115 \tan 80^\circ$$

In $\triangle DPC$

$$x^2 = DP^2 + PC^2 - 2 \times DP \times PC \cos P$$

$$x^2 = (115 \tan 67^\circ)^2 + (115 \tan 80^\circ)^2 - 2(115 \tan 67^\circ)(115 \tan 80^\circ) \cos 85^\circ$$

$$x = 684.076 \dots$$

mark.

3mk correct soln

2mk

$$-5 \leq x \leq -3$$

$$\text{or } x \geq -2$$

1mk working

correctly towards

soln.

3mk correct soln

2mk

• Finds a correct

expression

for DC^2

1mk

• Finds DP and PC

• finds DC^2

with incorrect

value/s.

Question 13.

a) i) LHS = $\sin A \cos B + \cos A \sin B$

$$+ \sin A \cos B - \cos A \sin B$$

$$= 2 \sin A \cos B$$

$$= \text{RHS}$$

1mk correct soln

(ii) $\int_0^{\frac{\pi}{6}} \sin 4x \cos 2x \, dx$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{2} (\sin 6x + \sin 2x) \, dx$$

$$= \frac{1}{2} \left[-\frac{1}{6} \cos 6x - \frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{6}}$$

$$= \frac{1}{2} \left[\left(-\frac{1}{6} \cos \frac{\pi}{2} - \frac{1}{2} \cos \frac{\pi}{3} \right) - \left(-\frac{1}{6} \cos 0 - \frac{1}{2} \cos 0 \right) \right]$$

$$= \frac{7}{24}$$

2mk - correct soln

1mk using (i).

b) $4^n + 14$ is multiple of 6

(A) Show $n=1$ works

$$4^1 + 14 = 18 = 6 \times 3$$

$$\therefore \text{true for } n=1$$

3mk - correct soln

2mk

• Proves C and, two of ABD.

(B) Assume true for $n=k$

$$\text{ie } 4^k + 14 = 6Q \text{ where } Q \text{ is a positive integer}$$

$$\text{ie } 4^k = 6Q - 14$$

1mk

• Proves C and one of ABD

• Shows ABD

(C) Test $n=k+1$

Assm: $4^{k+1} + 14 = 6P$ where P is a positive integer

$$\text{LHS} = 4 \times 4^k + 14$$

$$= 4(6Q - 14) + 14 \text{ (if assumption is true)}$$

Q13 b) cont.

$$\begin{aligned}
 &= 24Q - 42 \\
 &= 6(4Q - 7) \\
 &= 6P \text{ where } P = 4Q - 7 \\
 &\therefore \text{divisible by 6.}
 \end{aligned}$$

Q. If it is true for $n=k$, it is now proved for $n=k+1$

Since it is true for $n=1$, then it is true for all positive integers of $n \geq 1$.

c) $x^3 + 2x^2 + 3x + 6 = 0.$

$$\begin{aligned}
 \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\
 &= \frac{3}{-6} \\
 &= -\frac{1}{2}.
 \end{aligned}$$

d) MATHEMATICS

(i) $\frac{11!}{2!2!2!} = 4989600$

(ii) vowels (AAE) be 1 entity

$$\begin{array}{c}
 8! \leftarrow \text{number of entities} \\
 \begin{array}{cc}
 2! & 2! \\
 \uparrow & \uparrow \\
 2Ts & 2Ts
 \end{array}
 \times
 \begin{array}{c}
 4! \leftarrow \text{arrangement of vowels} \\
 2! \\
 \uparrow \\
 2As
 \end{array}
 \end{array}$$

$\therefore \text{total} = 120960$

(iii) Place M's at beginning and end

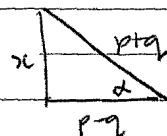
$$\begin{array}{c}
 9! \leftarrow \text{placing other letters} \\
 \begin{array}{cc}
 2! & 2! \\
 \uparrow & \uparrow \\
 2Ts & 2As
 \end{array}
 \end{array}$$

$\therefore \text{total} = 90720$

marks

Q13 e)

Let $\cos \alpha = \frac{p-q}{p+q}$



$$x^2 = (p+q)^2 - (p-q)^2$$

$$x^2 = 4pq$$

$$x = 2\sqrt{pq}$$

$$\sin(2\alpha) = 2\sin\alpha\cos\alpha$$

$$= 2 \times \frac{2\sqrt{pq}}{p+q} \times \frac{p-q}{p+q}$$

$$= \frac{4\sqrt{pq}(p-q)}{(p+q)^2}$$

marks

2mks - correct soln

link

- expands $\sin 2\alpha$
- uses triangle to find opposite side (x)

2mks - correct soln

link

- finds $\beta\gamma + \alpha\gamma + \alpha\beta$
- finds $\alpha\beta\gamma$

1mk - correct answer

2mk correct soln

link

- groups vowels as one entity

2mks correct soln

link

- placing M at beginning and end and arranging 9 other letters

Question 14.

a) $V = \pi \int y^2 dx$

(A) $\rightarrow V = \pi \int_0^{\frac{\pi}{3}} \sin^2 x dx$

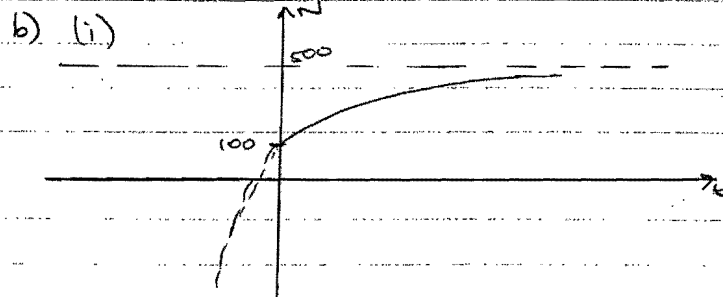
$V = \pi \int_0^{\frac{\pi}{3}} \frac{1}{2} \times (1 - \cos 2x) dx$

(B) $\rightarrow V = \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{3}}$

$V = \frac{\pi}{2} \left[\left(\frac{\pi}{3} - \frac{1}{2} \sin \frac{2\pi}{3} \right) - \left(0 - \frac{1}{2} \sin 2 \times 0 \right) \right]$

$V = \frac{\pi}{2} \left[\frac{\pi}{3} - \frac{1}{2} \times \frac{\sqrt{3}}{2} \right]$

$= \frac{\pi}{2} \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right]$



(ii) $N = 500 - 400(e^{-0.1t})$ then $400e^{-0.1t} = 500 - N$

$\frac{dN}{dt} = 0.1 \times 400 e^{-0.1t}$
 $= 0.1(500 - N)$

(iii) rate of growth = $\frac{1}{2} \times (\text{initial rate of growth})$
 initial rate of growth.

$= 0.1(500 - 100) = 0.1 \times 400$

population for $\frac{1}{2}$ rate of growth.

$= 0.1(500 - N) = (0.1 \times 400) \times \frac{1}{2}$

$N = 300$

marks

3marks - correct soln

2marks

• Successfully integrates $\sin^2 x$

1mk

• Correct application of volume formula ie (A).

2marks

(A) correct curve

(B) correct asymptote

(C) labels intercept

1mk

• Shows 2 out of 3 of ABC.

1mk - correct soln

1mk - correct answer

Q14c) (i) PQ : QS = -4 : 3

$Q = \left(\frac{-4 \times 0 + 3 \times 2ap}{-4+3}, \frac{-4 \times a + 3 \times ap^2}{-4+3} \right)$

$= (-6ap, 4a - 3ap^2)$

$= (-6ap, -a(4 - 3p^2))$

(ii) $x = -6ap$ $y = a(4 - 3p^2)$

$p = \frac{x}{-6a}$

Sub p into y.

$y = a \left(4 - 3 \times \left(\frac{-x}{6a} \right)^2 \right)$

$y = a \left(4 - 3 \times \frac{x^2}{36a^2} \right)$

$y = 4a - \frac{x^2}{12a^2}$

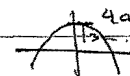
$x^2 = -12a(y - 4a)$

$4 \times \text{focal length} = 12a$

focal length = $3a$.

vertex $(0, 4a)$.

$\therefore \text{focus} = (0, a)$



marks.

2marks - correct soln

1mk

• uses internal formula.
 • does not show subst. of values into the formula.

2marks - correct soln

1mk

• Finds correct cartesian eqn.
 • From incorrect cartesian finds the focus.

$$OP = OQ \text{ (radii)}$$

Q14d) (i) $AP = AQ$ (tangents from an external pt are =)] OR marks - correct soln

$$\angle OQA = 90^\circ \text{ (tangent } \perp \text{ radius)}$$

$$\angle OPA = 90^\circ \text{ (tangent } \perp \text{ radius)}$$

$$\angle POQ + \angle OQA + \angle OPA + \angle PAQ = 360^\circ \text{ (Sum of quad)}$$

$$\therefore \angle POQ = 90^\circ$$

$\therefore AOPQ$ is a square (all angles 90°)

and one pair of adjacent sides =)

Link - shows

significant progress towards soln

(ii) Prove: $PA = \frac{1}{2}(c+b-a)$ ①.

marks - correct soln

Proof! $PA = AQ = PO$ (sides of a square)
= radius of circle, r .

Link shows

significant progress towards soln.

$$BP = c - r$$

$$QC = b - r$$

$$BR = BP = c - r \text{ (tangents from external pts are =)}$$

$$RC = QC = b - r \text{ (tangents from external pt are =)}$$

$$BC = c - r + b - r = a \text{ ———— ②}$$

$$\textcircled{1} \therefore \text{LHS} = PA = r$$

$$\text{RHS} = \frac{1}{2}(c+b-a)$$

$$= \frac{1}{2}(c+b - [c-r+b-r]) \rightarrow \text{using } \textcircled{2}.$$

$$= \frac{1}{2}(2r)$$

$$= r$$

$$= PA.$$

$$= \text{LHS}.$$

$$\therefore \text{LHS} = \text{RHS}.$$